A3: Teach - Group 29

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3D-model of beam and column: click here to open the model

1 Introduction

This is a tutorial on how geometry is represented in 3D-models. When working in Revit or other CAD/BIM programs, most users only see the finished object, such as a beam or a column, without knowing what the program actually creates behind the scenes. In practice, every placed element is stored as geometry defined by coordinate points (vertices). These points are connected by edges and faces to form a mesh, which together makes up the shape of the object.

The tutorial explains what this mesh consists of, how the coordinate points define an element's shape, position and orientation, and how this information is used for our tool.

2 How we came by the concept

When we started making a tool that was to automatically naming structural elements based on their placement, type and level, which would make it easier for tasks like vertical load distribution calculations. The first step of making this tool was to extract the center coordinates of each individual beam and plot the beams in python. When doing this it was discovered that the geometry of for example beams was made of coordinate points that together form an object, which we also could use to calculate the orientation of the object.

3 Tutorial

Geometrical IFC objects such as beams and columns are made of three geometric building blocks:

1. **Vertices (coordinate points):** Each vertex is a point in 3D space defined by an X, Y and Z coordinate, for example (0, 0, 0) or (0.3, 0, 3.0). These points are the foundation of the geometry.
2. **Edges:** A line between two vertices. Edges form the outline of the object.
3. **Faces:** A surface bounded by three or more edges. In a mesh, faces are typically triangles (or sometimes quadrilaterals), and many faces together form the closed surface of the object.

Together, vertices, edges and faces form a **mesh**. A simple rectangular beam, for example, can be described by 8 vertices (the corners), 12 edges and 6 faces (or 12 triangles, if we use triangular meshes). The more complex the shape, the more vertices and faces are needed, and curved surfaces such as a round column are approximated by many small flat faces.

4 Applicability

When the vertices of a beam is extracted, the orientation of the beam can be calculated by using singular value decomposition (SVD). The math behind this is somewhat complicated, but it basically takes all the points in the model, and defines the long direction of the beam. It takes the axis, where the points are most spread out, which for regular straight beams, are the longitudinal direction. By this method it is possible to extract the start and end points of all regular straight beams.